

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

Wednesday, 5 August 2026

Contents

1	Integrals	2
1.1	Medium Integral	2
1.2	Hard Integral	2
2	Differentials	4
2.1	Medium Differential	4
2.2	Hard Differential	4
3	Derivatives	6
3.1	Medium Derivative	6
3.2	Hard Derivative	6
4	Physics & Engineering	8
4.1	Electromagnetism	8
4.2	Quantum Mechanics	8
4.3	Mechanics	9

Integrals

1.1 Medium Integral

Gaussian Integral Difference

Evaluate the integral expression:

$$\int_{-\infty}^{\infty} (x+1)^2 e^{-\pi x^2} dx - \left(\int_{-\infty}^{\infty} (x+1) e^{-\pi x^2} dx \right)^2$$

Solution: We utilize standard results for Gaussian integrals over the entire real line:

$$\int_{-\infty}^{\infty} e^{-\pi x^2} dx = 1, \quad \int_{-\infty}^{\infty} x e^{-\pi x^2} dx = 0, \quad \int_{-\infty}^{\infty} x^2 e^{-\pi x^2} dx = \frac{1}{2\pi}$$

First, evaluate the second integral in the given expression:

$$\int_{-\infty}^{\infty} (x+1) e^{-\pi x^2} dx = \int_{-\infty}^{\infty} x e^{-\pi x^2} dx + \int_{-\infty}^{\infty} e^{-\pi x^2} dx = 0 + 1 = 1$$

Squaring this result gives $1^2 = 1$. Next, evaluate the first integral by expanding the integrand:

$$\int_{-\infty}^{\infty} (x^2 + 2x + 1) e^{-\pi x^2} dx = \int_{-\infty}^{\infty} x^2 e^{-\pi x^2} dx + 2 \int_{-\infty}^{\infty} x e^{-\pi x^2} dx + \int_{-\infty}^{\infty} e^{-\pi x^2} dx$$

Substituting the known Gaussian integral values:

$$= \frac{1}{2\pi} + 2(0) + 1 = 1 + \frac{1}{2\pi}$$

Finally, subtract the squared result of the second integral from the first:

$$\left(1 + \frac{1}{2\pi} \right) - 1 = \frac{1}{2\pi}$$

1.2 Hard Integral

Logarithmic Rational Power Integral

Evaluate the definite integral:

$$\int_0^{\infty} \frac{x^{1/4} \ln\left(\frac{x}{x+1}\right)}{(x+1)^2} dx$$

Solution: Let $u = \frac{x}{x+1}$. Then $x = \frac{u}{1-u}$ and $dx = \frac{1}{(1-u)^2} du$. The integration limits change as follows: when $x \rightarrow 0$, $u \rightarrow 0$, and when $x \rightarrow \infty$, $u \rightarrow 1$. Substitute these into

the integral:

$$\int_0^1 \frac{\left(\frac{u}{1-u}\right)^{1/4} \ln(u)}{\left(\frac{1}{1-u}\right)^2} \frac{1}{(1-u)^2} du = \int_0^1 u^{1/4} (1-u)^{-1/4} \ln(u) du$$

This integral can be interpreted through the derivative of the Beta function. Recall the definition of the Beta function:

$$B(a, b) = \int_0^1 u^{a-1} (1-u)^{b-1} du$$

Differentiating with respect to a brings down a $\ln(u)$:

$$\frac{\partial}{\partial a} B(a, b) = \int_0^1 u^{a-1} (1-u)^{b-1} \ln(u) du$$

Matching our integral, we set $a - 1 = \frac{1}{4} \implies a = \frac{5}{4}$ and $b - 1 = -\frac{1}{4} \implies b = \frac{3}{4}$. The derivative of the Beta function is given by:

$$\frac{\partial}{\partial a} B(a, b) = B(a, b) (\psi(a) - \psi(a+b))$$

where $\psi(z)$ is the digamma function. First, calculate $B\left(\frac{5}{4}, \frac{3}{4}\right)$:

$$B\left(\frac{5}{4}, \frac{3}{4}\right) = \frac{\Gamma(5/4)\Gamma(3/4)}{\Gamma(2)} = \frac{1}{4}\Gamma(1/4)\Gamma(3/4) = \frac{1}{4} \frac{\pi}{\sin(\pi/4)} = \frac{\pi\sqrt{2}}{4}$$

Next, compute the digamma terms $\psi(5/4) - \psi(2)$. Using recurrence relations $\psi(z+1) = \psi(z) + 1/z$:

$$\psi(5/4) = \psi(1/4) + 4 = -\gamma - \frac{\pi}{2} - 3\ln(2) + 4$$

$$\psi(2) = \psi(1) + 1 = -\gamma + 1$$

Subtracting the two yields:

$$\psi(5/4) - \psi(2) = 3 - \frac{\pi}{2} - 3\ln(2)$$

The final value of the integral is the product:

$$\frac{\pi\sqrt{2}}{4} \left(3 - \frac{\pi}{2} - 3\ln(2)\right)$$

Differentials

2.1 Medium Differential

Infinite System of Differential Equations

Solve the system of infinitely many differential equations:

$$2^{n-1} \frac{dx_n}{dt} = \sum_{k=1}^n x_k - \sum_{j=1}^{n-1} e^{2^{1-j}t}$$

Given $x_n(0) = 1$ for all $n \in \mathbb{N}$, find the value of $\sum_{n=1}^{\infty} \ln(x_n(n))$.

Solution: We solve for the first few terms to identify a pattern. For $n = 1$:

$$\frac{dx_1}{dt} = x_1 \implies x_1(t) = Ce^t$$

Applying $x_1(0) = 1$, we get $x_1(t) = e^t$. For $n = 2$:

$$2 \frac{dx_2}{dt} = x_1 + x_2 - e^t = e^t + x_2 - e^t = x_2 \implies \frac{dx_2}{dt} = \frac{1}{2}x_2 \implies x_2(t) = e^{t/2}$$

For $n = 3$:

$$4 \frac{dx_3}{dt} = x_1 + x_2 + x_3 - (e^t + e^{t/2}) = e^t + e^{t/2} + x_3 - e^t - e^{t/2} = x_3 \implies x_3(t) = e^{t/4}$$

By induction, we clearly have $x_n(t) = e^{t/2^{n-1}}$. We need to evaluate the infinite sum:

$$\sum_{n=1}^{\infty} \ln(x_n(n))$$

Substituting $x_n(n) = e^{n/2^{n-1}}$ into the logarithm:

$$\sum_{n=1}^{\infty} \ln(e^{n/2^{n-1}}) = \sum_{n=1}^{\infty} \frac{n}{2^{n-1}}$$

This is a standard arithmetico-geometric series. Let $S = \sum_{n=1}^{\infty} nr^{n-1}$. For $|r| < 1$, $S = \frac{1}{(1-r)^2}$. Here, $r = 1/2$:

$$S = \frac{1}{(1-1/2)^2} = \frac{1}{(1/2)^2} = \frac{1}{1/4} = 4$$

2.2 Hard Differential

Exponential Coefficients Differential Equation

Given that $y'' + (1 + 3e^x)y' + (3e^x + 2e^{2x})y = e^x e^{e^x}$, $y(0) = 0$, and that $y'(0) = e^{-1}$, find $y(-1)$. (Assume continuity over the interval.)

Solution: We can factor the differential operator $(D^2 + (1 + 3e^x)D + (3e^x + 2e^{2x}))y = e^x e^{e^x}$. Consider the factorization $(D + e^x)(D + 1 + 2e^x)y$. Let's test it:

$$\begin{aligned}(D + e^x)(y' + (1 + 2e^x)y) &= (y'' + 2e^x y' + (1 + 2e^x)y') + e^x(y' + (1 + 2e^x)y) \\ &= y'' + (1 + 3e^x)y' + (3e^x + 2e^{2x})y\end{aligned}$$

This perfectly matches our ODE. We introduce a substitution $u = y' + (1 + 2e^x)y$. The equation becomes a first-order linear ODE:

$$u' + e^x u = e^x e^{e^x}$$

The integrating factor is $I(x) = e^{\int e^x dx} = e^{e^x}$. Multiplying both sides:

$$(ue^{e^x})' = e^x e^{2e^x} \implies ue^{e^x} = \int e^x e^{2e^x} dx = \frac{1}{2}e^{2e^x} + C$$

$$u(x) = \frac{1}{2}e^{e^x} + Ce^{-e^x}$$

Using the initial conditions $y(0) = 0$ and $y'(0) = e^{-1}$, we find $u(0)$:

$$u(0) = y'(0) + 3y(0) = e^{-1}$$

$$e^{-1} = \frac{1}{2}e + Ce^{-1} \implies C = 1 - \frac{e^2}{2}$$

Now we solve $y' + (1 + 2e^x)y = u(x) = \frac{1}{2}e^{e^x} + (1 - \frac{e^2}{2})e^{-e^x}$. The integrating factor is $J(x) = e^{\int (1+2e^x)dx} = e^{x+2e^x}$.

$$(ye^{x+2e^x})' = e^{x+2e^x} \left[\frac{1}{2}e^{e^x} + (1 - \frac{e^2}{2})e^{-e^x} \right] = \frac{1}{2}e^x e^{3e^x} + (1 - \frac{e^2}{2})e^x e^{e^x}$$

Integrating both sides:

$$ye^{x+2e^x} = \frac{1}{6}e^{3e^x} + (1 - \frac{e^2}{2})e^{e^x} + K$$

Using $y(0) = 0$:

$$0 = \frac{1}{6}e^3 + (1 - \frac{e^2}{2})e + K \implies K = \frac{1}{3}e^3 - e$$

Finally, evaluating $y(x)$ at $x = -1$:

$$y(-1) = e^{1-2e^{-1}} \left(\frac{1}{6}e^{3e^{-1}} + \left(1 - \frac{e^2}{2}\right)e^{e^{-1}} + \frac{1}{3}e^3 - e \right)$$

Derivatives

3.1 Medium Derivative

Infinite Series Function Derivative

Let $f(x) = \sum_{n=1}^{\infty} \frac{(-1)^n (x^3 + 3x^2 + 3x)}{nx^2}$ for $x \neq 0$. Find $f'(\sqrt{2})$.

Solution: First, we factor out the x terms from the infinite sum since they do not depend on the summation index n :

$$f(x) = \left(\frac{x^3 + 3x^2 + 3x}{x^2} \right) \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

Simplify the rational expression by dividing by x^2 :

$$\frac{x^3 + 3x^2 + 3x}{x^2} = x + 3 + \frac{3}{x}$$

The infinite series is the standard Maclaurin expansion for $\ln(1+z)$ evaluated at $z = 1$, but negated:

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n} = - \left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots \right) = -\ln(2)$$

Substituting this back into the function:

$$f(x) = -\ln(2) \left(x + 3 + \frac{3}{x} \right)$$

Now, take the derivative of $f(x)$ with respect to x :

$$f'(x) = -\ln(2) \left(1 - \frac{3}{x^2} \right)$$

Finally, substitute $x = \sqrt{2}$:

$$f'(\sqrt{2}) = -\ln(2) \left(1 - \frac{3}{(\sqrt{2})^2} \right) = -\ln(2) \left(1 - \frac{3}{2} \right) = -\ln(2) \left(-\frac{1}{2} \right) = \frac{\ln(2)}{2}$$

3.2 Hard Derivative

Infinite Product Function Second Derivative

Let $f(x) = \prod_{n=1}^{\infty} \left(1 + \frac{x}{n} \right) e^{-x/n}$. Find $f''\left(\frac{1}{2}\right)$.

Solution: The given infinite product is intimately related to the Weierstrass factorization of the Gamma function $\Gamma(x)$:

$$\frac{1}{\Gamma(x)} = x e^{\gamma x} \prod_{n=1}^{\infty} \left(1 + \frac{x}{n} \right) e^{-x/n}$$

Where γ is the Euler-Mascheroni constant. Rearranging this gives an expression for $f(x)$:

$$f(x) = \frac{e^{-\gamma x}}{x\Gamma(x)} = \frac{e^{-\gamma x}}{\Gamma(x+1)}$$

To find the second derivative, we take the natural logarithm of both sides:

$$\ln f(x) = -\gamma x - \ln \Gamma(x+1)$$

Differentiating once with respect to x yields the digamma function $\psi(z) = \frac{\Gamma'(z)}{\Gamma(z)}$:

$$\frac{f'(x)}{f(x)} = -\gamma - \psi(x+1)$$

Differentiating a second time yields the trigamma function $\psi_1(z) = \psi'(z)$:

$$\frac{f''(x)f(x) - (f'(x))^2}{(f(x))^2} = -\psi_1(x+1)$$

$$f''(x) = f(x) \left[\left(\frac{f'(x)}{f(x)} \right)^2 - \psi_1(x+1) \right] = f(x) \left[(-\gamma - \psi(x+1))^2 - \psi_1(x+1) \right]$$

Evaluate the required components at $x = 1/2$. First, $f(1/2)$:

$$f(1/2) = \frac{e^{-\gamma/2}}{\Gamma(3/2)} = \frac{2e^{-\gamma/2}}{\sqrt{\pi}}$$

Next, evaluate the digamma and trigamma functions at $3/2$:

$$\psi(3/2) = 2 - \gamma - 2\ln(2) \implies -\gamma - \psi(3/2) = 2\ln(2) - 2$$

$$\psi_1(3/2) = \frac{\pi^2}{2} - 4$$

Substitute these back into the $f''(x)$ expression:

$$f''(1/2) = \frac{2e^{-\gamma/2}}{\sqrt{\pi}} \left[(2\ln(2) - 2)^2 - \left(\frac{\pi^2}{2} - 4 \right) \right]$$

$$f''(1/2) = \frac{2e^{-\gamma/2}}{\sqrt{\pi}} \left[4(\ln(2) - 1)^2 - \frac{\pi^2}{2} + 4 \right]$$

Physics & Engineering

4.1 Electromagnetism

Cylindrical Dielectric Force

A cylindrical layer of dielectric with relative permittivity $\varepsilon = 7$ is inserted into a cylindrical capacitor to fill the space between the electrodes. The mean radius is $R = 6.0$ cm, the gap is $d = 1.0$ mm, and $d \ll R$. A constant voltage $V = 1$ kV is applied. Find the magnitude of the electric force (in N) pulling the dielectric into the capacitor. Take $\varepsilon_0 = \frac{1}{36\pi \times 10^9} \text{ Fm}^{-1}$.

Solution: Let x be the length of the dielectric inserted into the capacitor, and L be the total length of the capacitor. The capacitor can be modeled as two capacitors in parallel: one filled with the dielectric (length x) and one with air (length $L - x$). Since $d \ll R$, the cylindrical capacitance formula $C = \frac{2\pi\varepsilon L}{\ln((R+d/2)/(R-d/2))}$ simplifies to a parallel-plate equivalent: $C \approx \frac{\varepsilon_0 A}{d} = \frac{\varepsilon_0 (2\pi R)L}{d}$. The total capacitance as a function of x is:

$$C(x) = \frac{2\pi\varepsilon_0 R}{d}(L - x) + \frac{2\pi\varepsilon_0 \varepsilon R}{d}x = \frac{2\pi\varepsilon_0 R}{d}[L + (\varepsilon - 1)x]$$

The energy stored in the capacitor at a constant voltage V is $U = \frac{1}{2}CV^2$. The electric force pulling the dielectric is the derivative of the energy with respect to the displacement x :

$$\begin{aligned} F &= \frac{\partial U}{\partial x} = \frac{1}{2}V^2 \frac{dC}{dx} \\ \frac{dC}{dx} &= \frac{2\pi\varepsilon_0 R(\varepsilon - 1)}{d} \\ F &= \frac{1}{2}V^2 \frac{2\pi\varepsilon_0 R(\varepsilon - 1)}{d} = \frac{\pi\varepsilon_0 R(\varepsilon - 1)V^2}{d} \end{aligned}$$

Now, substitute the given values: $V = 1000$ V, $R = 0.06$ m, $d = 0.001$ m, $\varepsilon = 7$, and $\varepsilon_0 = \frac{1}{36\pi \times 10^9}$.

$$\begin{aligned} F &= \frac{\pi \left(\frac{1}{36\pi \times 10^9} \right) (0.06)(7 - 1)(1000)^2}{0.001} \\ F &= \frac{\frac{1}{36 \times 10^9} \cdot 0.06 \cdot 6 \cdot 10^6}{10^{-3}} = \frac{0.36 \times 10^6}{36 \times 10^9 \times 10^{-3}} = \frac{0.36 \times 10^6}{36 \times 10^6} = \frac{0.36}{36} = 0.01 \text{ N} \end{aligned}$$

4.2 Quantum Mechanics

Fresnel Diffraction Intensity

A monochromatic plane wave of wavelength $\lambda = 632.8$ nm is incident on a perfectly opaque circular disc. For a point in the Fresnel diffraction pattern, let $C(u) = \int_0^u \cos(\frac{1}{2}\pi t^2) dt$ and $S(u) = \int_0^u \sin(\frac{1}{2}\pi t^2) dt$. The normalised intensity is $\frac{I}{I_0} = (1 - C(u))^2 + (1 - S(u))^2$. If $u = 1.500$, calculate the value of $100(I/I_0)$.

Give your answer to three significant figures.

Solution: The functions $C(u)$ and $S(u)$ are the standard Fresnel integrals. Evaluating them at $u = 1.500$ requires table lookups or numerical integration tools standard in optics calculations:

$$C(1.500) \approx 0.44526, \quad S(1.500) \approx 0.69750$$

Substitute these values into the normalized intensity equation:

$$\frac{I}{I_0} = (1 - 0.44526)^2 + (1 - 0.69750)^2$$

$$\frac{I}{I_0} = (0.55474)^2 + (0.30250)^2 \approx 0.3077 + 0.0915 = 0.3992$$

To find the required quantity, we multiply by 100:

$$100 \left(\frac{I}{I_0} \right) = 100 \times 0.3992 = 39.92$$

Rounding to three significant figures gives 39.9.

4.3 Mechanics

Relative Satellite Periods

A geostationary satellite orbits Earth at a distance r_1 from the center. A second satellite orbits in the equatorial plane in the opposite direction to Earth's rotation at a distance r_2 . Given $r_1 = 1.21r_2$ and the time period of the second satellite as measured from the first is $24/p$ hours, find the value of p .

Solution: By Kepler's Third Law, the square of the orbital period is proportional to the cube of the orbital radius ($T^2 \propto r^3$). For the geostationary satellite, $T_1 = 24$ hours. For the second satellite:

$$\left(\frac{T_2}{T_1} \right)^2 = \left(\frac{r_2}{r_1} \right)^3$$

We are given $r_1 = 1.21r_2 \implies \frac{r_2}{r_1} = \frac{1}{1.21} = \frac{1}{1.1^2}$. Substituting this in:

$$\left(\frac{T_2}{T_1} \right)^2 = \left(\frac{1}{1.1^2} \right)^3 = \frac{1}{1.1^6} \implies \frac{T_2}{T_1} = \frac{1}{1.1^3} = \frac{1}{1.331}$$

$$T_2 = \frac{24}{1.331} \text{ hours}$$

The angular velocities of the two satellites are $\omega_1 = \frac{2\pi}{T_1}$ and $\omega_2 = \frac{2\pi}{T_2}$. Since they orbit in opposite directions, their relative angular velocity is the sum:

$$\omega_{rel} = \omega_1 + \omega_2 = \frac{2\pi}{T_1} + \frac{2\pi}{T_2}$$

The relative time period T_{rel} as observed from the first satellite is:

$$T_{rel} = \frac{2\pi}{\omega_{rel}} = \frac{1}{\frac{1}{T_1} + \frac{1}{T_2}} = \frac{T_1 T_2}{T_1 + T_2}$$

Substitute $T_1 = 24$ and $T_2 = 24/1.331$:

$$T_{rel} = \frac{24 \left(\frac{24}{1.331} \right)}{24 + \frac{24}{1.331}} = \frac{\frac{24}{1.331}}{1 + \frac{1}{1.331}} = \frac{24}{1.331 + 1} = \frac{24}{2.331} \text{ hours}$$

We are given $T_{rel} = 24/p$. Thus, $p = 2.331$.